DO NOW — RECALL

Lesson 103 · Input and Output: Selecting Devices for Ta

Name: _____ Class: _____ Date: _____

1. What does an input device do?

2. What does an output device do?

3. What is an actuator?

4. A 2×3 display grid: how many cells?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Captures user/environment data.

2. Model answer

Communicates information or causes action.

3. Model answer

Output device controlling physical movement/action.

4. Model answer

Six.

By the end, you can…

• An input device captures data or a user action; an output device communicates information or changes physical conditions.

• A temperature sensor measures physical conditions; an actuator controls a physical action; a display shows values to the user.

• Choose devices by required accuracy, accessibility, environment and response time. A touchscreen can perform both input and output functions.

Vocabulary — know these terms

input — Data entering computer system.

output — Information/effect produced.

sensor — Measures a physical quantity.

actuator — Creates a physical action.

NEW LEARNING

Read, annotate and explain

Understand

An input device captures data or a user action; an output device communicates information or changes physical conditions.

Apply

A temperature sensor measures physical conditions; an actuator controls a physical action; a display shows values to the user.

Evaluate

Choose devices by required accuracy, accessibility, environment and response time. A touchscreen can perform both input and output functions.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Recommend input and output for automated greenhouse temperature control.

Explain the choice

Input thermistor/temperature sensor; output fan/heater actuator, possibly a display.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For a room-temperature controller, choose appropriate input and output devices and explain roles. [6]

Question 2

Compare barcode reader and RFID for counting library books, linking benefits to context. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Design accessible touchscreen ticket kiosk: give two input and two output features and justify. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A touchscreen can only be an output device.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.